

1. Pre-training: what a base LLM actually learns

The equation on the early slide is the **pre-training loss**:

$$\mathcal{L}_{\text{pretrain}} = -\sum_t \log P(x_t \mid x_{<t})$$

$$\mathcal{L}_{\text{pretrain}} = -\sum \log P(x_t \mid x_{<t})$$

In plain words:

- The model sees a **sequence of tokens** (words, subwords, punctuation).
- At each position (t), it tries to **predict the next token** (x_t) given all previous tokens ($x_{<t}$).
- The loss is the **negative log-likelihood**, i.e. “how surprised am I by the real next token?” summed over all positions. Lower is better.

So a base LLM is literally trained to be a **next-token predictor** over massive internet text.

Example: “What is the capital of France”

Slide: “What will the base LLM output for this? What is the capital of France” then options:

- A: “What is its population?”
- B: “Paris”

A pure pre-trained model (no instruction tuning) might actually think:

“On the internet, after a question like this, it’s common to see *another* follow-up question or some continuation of discussion.”

So it might output something like **A** (another question) instead of answering with “Paris”.

Key idea:

- **Base model = document completer.** It just continues text in whatever style it saw during pre-training, not “answering the user” on purpose.
-

2. Paradigm shift: from “document completer” to “assistant”

Slide: *Paradigm Shift*

- **Pre-training:** massive text, next-token prediction (NTP).
- **Post-training:** instruction tuning + preference optimization on top.

Terms:

1. Base Model

- Learned just from unlabeled text.
- Good at language, facts, patterns.
- But not aligned with “be helpful, honest, harmless”.

2. Instruct / Chat Model

We take the base model and **fine-tune it** so that when the input looks like:

```
{"role": "user", "content": "Explain X"}
```

- it behaves like a **helpful assistant**.
- This is the “ChatGPT-style” model.

So the paradigm shift is:

From “predict what comes next in generic text” → “follow instructions, answer questions, chat, respect user preferences and safety”.

3. Supervised Fine-Tuning (SFT) / Instruction Tuning

Slide: *Supervised Fine-tuning (SFT) → Behavior Cloning*

What happens?

We build a dataset of conversation examples like:

```
{
  "messages": [
    {"role": "user", "content": "How do I bake a pizza?"},
    {"role": "assistant", "content": "Step 1 ... Step 2 ..."}
  ]
}
```

- Each pair is one training example: **input = all earlier messages, target = next assistant message.**
- We fine-tune the base model on millions of such pairs.

Why “behavior cloning”?

Because the model is **not** learning “why” an answer is good, or which answer is *better* than others.

It just learns:

“Given this user message, humans tend to respond *like this*. I'll mimic that pattern.”

Key points you can write in an exam:

1. **SFT = imitate human answers** on supervised (input, output) pairs.
2. It turns a base model into an **instruction-following chat model**.
3. But it **doesn't know which answer is better**; it only copies style and content of the training responses.

Slide with pizza example: they show possible “correct behaviors”:

- Asking clarification (family of six?).

- Asking follow-up questions (ingredients, time).
- Actually giving step-by-step instructions.

All of those are learned via **behavior cloning**: the model copies what humans did in the dataset.

4. Preference Fine-Tuning: why we need RLHF

Slide: *Preference-finetuning*

Problems:

- Users ask about **controversial / sensitive topics**: politics, religion, drugs, etc.
- It's hard to define perfect rules.
- Too strict → boring, “sorry I can’t answer anything”.
- Too loose → offensive / unsafe content.

We need a way to tell the model:

“Among several valid answers, some are **preferred** by the company / society / policy.”

This is where **RLHF (Reinforcement Learning from Human Feedback)** comes in.

5. RLHF Pipeline

Slide: *RLHF Pipeline aka why ChatGPT became successful* + diagram.

Three main stages:

5.1 Stage 1 – SFT (we already saw)

- Pretrained base model → **SFT model** using labeled chat data.

5.2 Stage 2 – Reward Modeling

We create a dataset of **comparisons**, not just single answers:

- For a given prompt, humans see **two or more model outputs** and rank them:
 - Response A is better than B (more helpful, safer, etc.).

We train a **Reward Model**:

- Input: (prompt, response)
- Output: a **scalar score** “how good is this response”.

This is basically a regressor that learns human preferences:

“Given this prompt and answer, humans rated it 4/5 vs 1/5.”

5.3 Stage 3 – RL Optimization (PPO)

Now we treat the SFT model as a **policy** in reinforcement learning:

- **Policy**: the model chooses a sequence of tokens (its answer).
- **Reward**: we feed (prompt, answer) into the **Reward Model**.

Objective:

Adjust the policy (the chat model) so that it produces responses with **higher reward scores**.

Algorithm: **PPO (Proximal Policy Optimization)**

- A standard RL algorithm that updates the policy but keeps each update **small / safe**, so the model doesn’t suddenly become weird.

Reward hacking / specification gaming

Because reward is only an **approximation of what we want**, the model can “hack” it:

- It might learn tricks that make the Reward Model happy but aren't actually good behavior (e.g., always apologizing, flattering, etc.).

To prevent the model from drifting too far from its original knowledge / style, we add a **KL Divergence penalty**:

- Encourage the final RLHF model to stay **close to the SFT/base model's distribution**.
- So the model can't completely "cheat" by moving into unnatural responses.

So an exam-style summary:

1. RLHF = SFT → Reward Model → RL (PPO) with KL penalty.
2. Reward model trained on **ranked human preferences**.
3. RL step updates model to maximize reward but must monitor **reward hacking**.

The monster diagram slide basically says:

- Big pretraining "beast" (huge unsupervised learning).
- Tiny pink patch = SFT.
- Even smaller = RLHF tweaks.

6. Automatic metrics for text: BLEU & ROUGE

Slides: "Detour; BLEU, ROUGE" etc.

6.1 BLEU – how to score a translation

Scenario:

- Original French: "J'ai mangé trois filberts." (or similar sentence)
- Reference translations (human):

- “I have eaten three hazelnuts.”
- “I ate three filberts.”

Model output: “I ate three hazelnuts.”

We want **one number** between 0 and 1 (or 0–100) that says “how good” the translation is, using only:

- the **reference(s)**
- the **model output**

Unigram precision (BLEU’s starting point)

- Look word by word in the model output.
- If a word appears in *any* reference → count as 1, else 0.
- Divide “matching words” / “total words” → value between 0 and 1.

Problem: the silly sentence “three three three three” has high unigram precision if “three” appears in references.

To fix this, BLEU:

1. **Caps count per word** by its max frequency in references. (So you can’t use “three” 10 times if it appears once.)
2. Uses **n-grams** (bigrams, trigrams, 4-grams), not just unigrams.
 - For example bigrams from “I ate three hazelnuts”:
 - “I ate”, “ate three”, “three hazelnuts”.

This punishes wrong word order like “ate hazelnuts I three”.

Brevity penalty

If the model produces a **very short** sentence like “I ate”, BLEU precision might be high (words all correct) but translation is incomplete.

So BLEU multiplies by a **brevity penalty**:

- If output length $\geq$ closest reference length $\rightarrow$ penalty = 1 (no change).
- If output shorter $\rightarrow$ penalty = $(e^{1 - (\text{ref length} / \text{output length})})$.
 - Bigger penalty if output is much shorter.

Pros of BLEU

- Fast & cheap to compute.
- Widely used $\rightarrow$ easy to compare across papers & models.

Cons of BLEU

- Doesn't understand **meaning** (semantics). It's purely surface n-grams.
- Sensitive to wording:
 - "I consumed the apple."
 - "I ate an apple."
Both are perfectly fine translations of "J'ai mangé la pomme", but BLEU can score them badly because they don't match exact words or articles.
- Struggles with **morphologically rich languages** (where one root word has many forms) because exact surface form changes.

6.2 ROUGE – more recall-focused, popular for summarization

Slide: "ROGUE" (actually ROUGE)

Variants:

- **ROUGE-N** – overlap of n-grams (unigrams, bigrams).
- **ROUGE-L** – Longest Common Subsequence (LCS); this captures word order better.

Main differences from BLEU:

- BLEU = **precision**: “How many of my output n-grams appear in the reference?”
- ROUGE = **recall**: “How many of the reference n-grams appear in my output?”

So ROUGE is often used for **summarization**:

- A good summary should cover most of the **important content** of the reference (high recall), even if it adds some extra words.
-

7. Other metrics (Et al.)

Slide “Et al..” lists several other metrics.

You can memorize short 1-line definitions:

1. **Perplexity**

- Used for language models.
- Measures how well model’s predicted distribution matches the real text.
- Lower perplexity = model is **less surprised**, i.e. better at predicting.

2. **WER (Word Error Rate)** – speech recognition

- Measures substitutions, deletions, insertions needed to turn model output into reference text.
- Lower WER = better.

3. **Subtree Metric (STM)**

- Compares syntactic parse trees for reference vs output.
- Penalizes different **syntax structures**, even if words are similar.

4. **METEOR**

- Like BLEU but: uses **stems, synonyms, paraphrases**.

- Designed more for **sentence-level** comparison.

5. TER (Translation Error Rate)

- Number of edits (insert, delete, substitute, reorder) to change model translation → acceptable human translation.
- Lower TER = fewer edits needed → better.

8. Entropy, Cross-Entropy, Perplexity (in more detail)

Slides: Entropy, Cross-entropy, Perplexity.

8.1 Entropy

- Measures **uncertainty** or **surprise** in a distribution.
- For language:
 - Low entropy: next word is very predictable.
 - “The cat sat on the ____” → “mat” is highly expected.
 - High entropy: many possible next words.

Mathematically, for distribution P :

$$[H(P) = -\sum_x P(x) \log P(x)]$$

- If P is very peaked (one token has probability ~ 1), entropy is low.
- If P is spread out, entropy is high.

8.2 Cross-entropy

Now imagine:

- **P = true data distribution** (real language).
- **Q = model's distribution.**

Cross-entropy ($H(P, Q)$):

“On average, how surprised is model Q when the world behaves like P?”

Formula:

$$[H(P, Q) = -\sum_x P(x) \log Q(x)]$$

- If Q matches P perfectly, cross-entropy = entropy $H(P)$.
- Difference between them is **KL divergence**:

$$[D_{\{KL\}}(P \parallel Q) = H(P, Q) - H(P)]$$

In training:

- We minimize cross-entropy (or equivalently maximize likelihood).
- Lower cross-entropy = model's distribution closer to real language.

8.3 Perplexity

Perplexity = exponential of cross-entropy.

Intuition:

“If the model had to guess, how many equally likely options does it *feel* like it has?”

- Perplexity 1 → model is perfectly confident (one option).
- Perplexity 100 → model is like “there are ~100 equally likely tokens” → very uncertain.

Perplexity is a simple single number to compare language models, but:

- It's a **proxy**. Good perplexity doesn't always mean great real-world performance.
-

9. Evaluation of LLMs: why it's hard

Slides "Evaluation", "First, the more intelligent AI models become, the harder it is to evaluate them", etc.

Two big problems:

1. Open-ended outputs

- For text generation, code, essays → there are **many valid answers**.
- You can't pre-write all correct outputs.
- So simple exact-match is not enough.

2. Limited knowledge of internals

- If model weights, data, architecture are private, you can only treat it as a **black box**: give input, get output, judge quality.

So evaluation becomes:

- Choosing tasks, datasets, and metrics that correlate with what we actually want (helpfulness, correctness, safety).
-

10. Benchmarks for LLMs

Slide listing StoryCloze, MATH, GLUE, SuperGLUE, NaturalInstructions, MMLU etc.

Two types:

1. Single-task benchmarks

- E.g., TruthfulQA (avoid common myths),

- MATH (competition math), etc.

2. Multi-task benchmarks

- **GLUE (2018)** – multiple NLP tasks (sentiment, entailment, etc.).
- **SuperGLUE (2019)** – harder version.
- **NaturalInstructions** – instruction-following benchmark.
- **MMLU (2020) & MMLU-Pro (2024)** – multi-subject exam questions (history, physics, medicine, etc.).

Earlier, for language models, we mostly compared **cross-entropy**, **perplexity**, **bits per character (BPC)**, **bits per byte (BPB)**, which capture predictive ability but not full “reasoning” and “safety”.

11. Functional correctness & reference-based evaluation

Slides: “Functional Correctness (open-ended)”, “Reference Data”.

11.1 Functional correctness

Especially for **code generation**:

- Rather than only checking if generated code **matches** reference code,
- We **run the code** on unit tests and see:
 - Does it produce correct outputs?
 - Does it crash / time out?

Example: **HumanEval (OpenAI)** – 164 programming problems:

- Prompt: problem description.
- Model outputs Python function.

- Score = % of problems for which generated solution passes tests.

This is called **execution accuracy** or “pass@k”.

11.2 Reference data evaluation

For translation, summarization, question answering etc.:

- You often have **ground-truth / canonical responses**.
- You compute similarity:
 - **Exact** (string match).
 - **Lexical** metrics: BLEU, ROUGE, METEOR, TER, CIDEr (n-gram overlap).
 - **Semantic** metrics: based on **embeddings** (e.g., BERTScore, MoverScore).

Semantic metrics:

- Convert texts into vectors (e.g., from BERT).
 - Compare similarity in embedding space (cosine similarity).
 - Better at handling synonyms and paraphrases.
-

12. LLM-as-a-Judge

Slides: “LLM as a Judge”, diagrams with PandaLM, LangChain, etc.

Why use another LLM as judge?

For large-scale evaluation:

- Human evaluation is slow and expensive.
- So we prompt a **strong LLM** (e.g., GPT-4) like:

“Here is the question. Here are two answers (A, B). Which one is better and why?”

This is called **AI feedback** or “LLM-as-a-Judge”.

The diagram shows:

- Human gives instruction + input.
- Different models (Bloom, LLaMA, etc.) generate responses.
- “Judge model” (PandaLM or GPT-4) evaluates and picks the better one, sometimes with explanation.

Comparative evaluation

Instead of numeric score, we ask:

- “Which of these two answers do you prefer?”

This is used both:

- In RLHF labelling (human or LLM doing pairwise preference).
- In research benchmarks comparing models.

The slide warns: this is **not suitable for Yes/No** type questions where there is a clear single correct answer. Also asking users to pick every time can be annoying (“math example” screenshot).

Self-preference bias & filter bubble

Important concept:

- If you use **GPT-4** as judge, it tends to prefer answers produced by **GPT-4-style models** (similar phrasing, structure) over others (e.g., Claude).
- So models get optimized to “sound like the judge”.

This creates a **filter bubble**:

- The whole ecosystem becomes biased toward one model’s style.

- Evaluation no longer purely measures truthfulness/helpfulness; it also checks “do you speak like GPT-4?”.
-

13. Task- and domain-specific evaluation

Slides near the end:

- “models are usually tested domain-wise”
- “exact evaluation”
- “coding → functional correctness”
- “efficiency → time taken for the SQL query, memory usage”
- “topical content → MCQs”
- “Dataset: TruthfulQA”

Idea: you don’t just want **one global score**; you care about **domain-specific skills**:

1. Exact evaluation

- For some tasks, there really is one correct answer:
 - e.g., arithmetic, SQL queries, classification labels.
- You can use exact string match or normalized match.

2. Coding

- Evaluate by **functional correctness**: run tests, measure pass rate.

3. Efficiency

- For generated SQL or algorithms: also check **runtime, memory usage**.
- Two correct solutions, but one is faster/more efficient → prefer that.

4. Topical content

- For knowledge tests: multiple-choice (MCQs), short answers.
- Example: medical, legal exam questions.

5. Truthfulness

- Datasets like **TruthfulQA** check whether model avoids **common false myths** and hallucinations.

14. Benchmarking caveats & Goodhart's Law

Slides: "Benchmarking", Goodhart's Law, then list of many benchmarks (MATH, LegalBench, MedQA, WMT, NarrativeQA, OpenBookQA, Natural Questions).

Goodhart's Law:

"When a measure becomes a target, it ceases to be a good measure."

For LLMs:

- Once everyone optimizes directly for a benchmark (e.g., MMLU, HumanEval), they start:
 - Overfitting on it.
 - Maybe even including parts of it in training data (data contamination).
- Then the benchmark no longer tells you real generalization; models "memorize" test.

The slide repeats the list of benchmarks to show there's a **zoo of specialized benchmarks**:

- For math: **MATH**
- Legal: **LegalBench**
- Medical: **MedQA**
- Translation: **WMT 2014**

- Reading comprehension: **NarrativeQA, OpenBookQA**
- General QA: **Natural Questions (with/without context)**

Key exam points:

1. Benchmarks are useful but can be **gamed**.
2. Need to keep updating / rotating benchmarks.
3. Need **robust evaluation** beyond simple leaderboards.

15. SelfCheckGPT: checking hallucinations via consistency

Final slide: *SelfCheckGPT*

Idea:

1. Ask the model the **same question 5 times** with **high temperature** (more randomness).
2. Compare the 5 answers:
 - For a factual question the model truly **knows** (e.g., “capital of France”):
 - All 5 answers will be essentially the same fact (“Paris”), even if wording differs.
 - For something it’s **hallucinating**:
 - The answers may disagree (different numbers, different events).

Assumption:

Consistency implies factuality (rough heuristic).
Divergence implies hallucination.

So SelfCheckGPT is a **meta-evaluation** technique:

- No external ground truth needed; uses the model's own stability as a signal of trustworthiness.

1. What even is an LLM doing?

Your first slide shows this formula:

$$\mathcal{L}_{\text{pretrain}} = -\sum \log P(x_t \mid x_{<t})$$

(Page 3)

$$\mathcal{L}_{\text{pretrain}} = -\sum \log P(x_t \mid x_{<t})$$

Forget the math for a moment.

This just means one very simple thing:

👉 A Large Language Model (LLM) is trained to **guess the next word**.

Example:

If I write:

The cat is sitting on the...

You naturally think: **mat**.

The model does the same thing. It looks at all previous words and guesses the next one.

So:

- It reads billions of sentences from the internet.
- For every sentence, it keeps guessing the next word.
- If it guesses wrong → it gets corrected.
- Over time, it becomes very good at guessing.

That's **pre-training**.

2. Why does a base LLM not always answer properly?

The slide asks:

What will the base LLM output for this?

What is the capital of France?

Options:

A) What is its population?

B) Paris

(Page 4–5)

A normal human would say: **Paris**.

But a **base LLM** might continue like a website article or a forum:

“What is its population?”

Why?

Because it is trained to **continue text**, not to act like a helpful assistant.

That’s why the slide says:

- Base Model = **Document Completer**
- Instruct Model = **Helpful Assistant**
(Page 6)

So:

- Base model = just continues text.
- Instruct model = actually answers you.

3. How do we turn it into a helpful assistant?

We use something called **Supervised Fine-Tuning (SFT)**.

Your slide says:

Supervised Fine-tuning (SFT) → Behavior Cloning
(Page 7)

Let me make this very simple:

They give the model **example conversations** like:

User: How do I bake a pizza?

Assistant: First preheat the oven to 200°C...

And the model learns:

“When humans talk like this, another human replies like that.”

This is called **behavior cloning** — it just copies how humans reply.

The slide even explains:

“Model isn’t learning *why* an answer is good, it just mimics the style.”

(Page 7)

4. Now: Why preference fine-tuning is needed

Your slide:

AI faces ethical challenges in controversial questions...

Excessive censorship makes model boring

(Page 10)

Problem:

Some questions are sensitive (politics, religion, violence, etc.).

Different people want different answers.

Some answers may hurt people.

So what do they do?

They make humans **compare answers** and say:

This answer is better than that one.

This is called **Preference Fine-Tuning**.

5. RLHF – The reason ChatGPT works well

Slide:

RLHF pipeline aka why ChatGPT became successful

(Page 12)

RLHF = **Reinforcement Learning with Human Feedback**

It has **3 steps**:

Step 1: SFT

We already did this. Teach model normal conversation.

Step 2: Reward Model

- Show humans 2 answers from AI.
- Humans rank them.
- A new AI model learns:
“Humans like this answer more than that one.”

This new model is called a **Reward Model**.

Step 3: Reinforcement Learning (PPO)

Now the main AI tries to make answers that:

- Get **high score** from the Reward Model.

But your slides warn:



Reward Hacking

The AI might try to cheat and say things it *knows* the reward model likes, not what’s actually correct.

So they add a **KL penalty** to stop the AI from becoming weird.

(Page 12–13)

6. BLEU – “How good is this translation?”

Now your slides talk about BLEU.

Example given:

French sentence:

J'ai mangé trois filberts.
(Page 16)

Human translations:

- I have eaten three hazelnuts.
- I ate three filberts.
(Page 17)

Model translates:

I ate three hazelnuts.
(Page 18)

Now question:

How do we give a **score** to this translation?

BLEU does:

1. Check how many words match the reference.
2. Check word order (using 1-grams, 2-grams, 3-grams, 4-grams).
3. Add **brevity penalty** if the sentence is too short.
(Page 21–27)

Problem examples from your slides:

- “Three three three three” (wrong but looks good to machine).
- “I ate.” (too short).
(Page 22, 26)

BLEU Pros:

- Fast
- Easy
- Widely used
(Page 28)

BLEU Cons:

- Doesn't understand meaning
 - "I consumed the apple" may get low score even if correct (Page 29–31)
-

7. ROUGE – Similar but for summaries

ROUGE focuses on:

👉 How **much of the important content** did you cover?

Your slide says:

- BLEU → precision
- ROUGE → recall (Page 34)

Simple meaning:

- BLEU: How many of *your* words match?
 - ROUGE: How many *important words* from the original you covered?
-

8. Entropy, Cross-Entropy & Perplexity (Super simple)

Your slides explain:

Entropy

"How predictable is the next word?"
(Page 42)

Example:

"The cat sat on the ____"

- If everyone guesses “mat” → low entropy
- If everyone guesses different words → high entropy

Cross Entropy

“How confused is the model compared to real language?”
(Page 43)

Perplexity

How many choices the model *feels* like it has.
(Page 44)

Simple idea:

- Perplexity = 1 → extremely confident
 - Perplexity = 100 → very confused
-

9. Why evaluating LLMs is hard

Your slide says:

The more intelligent AI becomes, the harder it is to evaluate
(Page 38)

Because:

- There are many possible right answers.
 - The model’s internal code and training are hidden.
(Page 39–40)
-

10. Benchmarks (Why so many tests?)

Your slide shows examples:

- MATH (for math problems)
- LegalBench (for law)
- MedQA (for medicine)
- TruthfulQA (for truthfulness)
(Page 41, 58, 60)

Each one checks a different skill.

But your slide also gives **Goodhart's Law**:

When a measure becomes a target, it stops being a good measure.
(Page 59)

Meaning:

If models train directly for a benchmark, they just learn that exam, not real intelligence.

11. LLM as a Judge

Your slides show one AI judging another AI.
(Page 47–52)

Problem:

GPT-4 will prefer GPT-4 style answers.

This is called: **Self-Preference Bias**.

It creates a **filter bubble** where all models start sounding the same.

12. SelfCheckGPT – How to check hallucination

Your final slide explains this clearly:
(Page 61)

Method:

1. Ask the model the **same question 5 times**.

2. Increase randomness.

3. Compare answers.

If all answers are similar → likely true.

If answers differ → likely hallucinating.

Example:

- “Capital of France” → always Paris 
- “Who caused event X?” → different answers 

1. What does the pre-training loss function shown in the lecture minimize?

- A) Difference between model output and reward signal
- B) Negative log likelihood of next token
- C) Reinforcement learning reward
- D) KL divergence between two models

✓ **Answer:** B

 **Explanation:** The pre-training objective minimizes the negative log probability of the correct next token as shown on page 3.

2. Why might a base LLM answer “What is its population?” instead of “Paris” for the capital of France?

- A) It is trained on QA datasets
- B) It performs semantic reasoning
- C) It behaves as a document continuator
- D) It uses reinforcement learning

✓ **Answer:** C

 **Explanation:** The lecture states that base models act as **document completers**, not assistants (Page 6).

3. What transformation happens during the “paradigm shift” mentioned in the slides?

- A) From speech model to vision model
- B) From code generator to chatbot
- C) From document completer to helpful assistant
- D) From weak AI to AGI

✓ **Answer:** C

 **Explanation:** This exact phrase appears in the paradigm shift slide (Page 6).

4. Why is SFT called "behavior cloning"?

- A) Because it copies network weights
- B) Because it learns reasoning strategies

- C) Because it mimics human answer styles
- D) Because it reverses reinforcement learning

✓ **Answer: C**

 **Explanation:** Slide directly says model mimics human answer writing styles, not reasoning (Page 7).

5. Which example did the slide use to show different valid behavioral responses?

- A) Capital of France
- B) Solving integrals
- C) Pizza making question
- D) Translation question

✓ **Answer: C**

 **Explanation:** The pizza example demonstrating behavior cloning is on Page 8.

6. Why do we need preference fine-tuning?

- A) To reduce training cost
- B) To avoid expensive GPUs
- C) To handle controversial topics and value trade-offs
- D) To increase speed of inference

✓ **Answer: C**

 **Explanation:** Preference fine-tuning is introduced specifically for handling controversial topics (Page 10).

7. Which is NOT one of the three stages of RLHF?

- A) SFT
- B) Reward modeling
- C) PPO Optimization
- D) Token pruning

✓ **Answer: D**

 **Explanation:** Token pruning is never mentioned; the pipeline is SFT → Reward Modeling → RL (Page 12).

8. What is "reward hacking"?

- A) When humans give wrong labels
- B) When model learns to exploit weaknesses in reward model
- C) When GPU overheats
- D) When model hallucinates answers

✓ **Answer:** B

 **Explanation:** The slide warns models can game the reward model (Page 12).

9. Why is KL divergence penalty added during RLHF?

- A) To speed up learning
- B) To improve translation quality
- C) To prevent model drifting too far from base model
- D) To reduce dataset size

✓ **Answer:** C

 **Explanation:** KL penalty keeps model behavior close to original distribution (Page 12).

10. BLEU score is primarily based on what?

- A) Semantic meaning
- B) Grammatical correctness
- C) N-gram overlaps
- D) Human opinions

✓ **Answer:** C

 **Explanation:** BLEU uses unigram to 4-gram precision (Page 25).

11. Why is the sentence "Three three three three" problematic for unigram precision?

- A) It contains no verbs
- B) It artificially inflates word match score
- C) It is grammatically incorrect
- D) It has no semantic meaning

✓ **Answer:** B

 **Explanation:** The slide explicitly shows this example (Page 22).

12. What does the brevity penalty in BLEU correct for?

- A) Long translations
- B) Repetition errors
- C) Translations that are too short
- D) Spelling mistakes

✓ **Answer:** C

 **Explanation:** Designed to punish overly short translations (Page 27).

13. BLEU fails badly on which type of languages?

- A) Isolated languages
- B) Agglutinative / Morphologically rich languages
- C) Dead languages
- D) Programming languages

✓ **Answer:** B

 **Explanation:** Slide explicitly states this problem (Page 33).

14. What does ROUGE emphasize over BLEU?

- A) Precision
- B) Recall
- C) Speed
- D) Tokenization

✓ **Answer:** B

 **Explanation:** ROUGE focuses on recall instead of precision (Page 34).

15. Which ROUGE variant uses longest common subsequence?

- A) ROUGE-N
- B) ROUGE-B

- C) ROUGE-L
- D) ROUGE-P

✓ **Answer:** C

 **Explanation:** ROUGE-L uses LCS (Page 34).

16. What does high entropy indicate?

- A) Highly predictable next word
- B) Low uncertainty
- C) High randomness or surprise
- D) Perfect model

✓ **Answer:** C

 **Explanation:** High entropy means unpredictable output (Page 42).

17. What does cross-entropy measure between P and Q?

- A) Difference between two datasets
- B) How well model predicts real language
- C) GPU efficiency
- D) Token compression

✓ **Answer:** B

 **Explanation:** It measures how surprised model Q is when reality follows P (Page 43).

18. Perplexity measures what intuitively?

- A) Model size
- B) Training time
- C) Model confusion / uncertainty
- D) Dataset quality

✓ **Answer:** C

 **Explanation:** Perplexity reflects how many likely guesses the model feels it has (Page 44).

19. Why is LLM evaluation difficult?

- A) Models are too small
- B) No GPUs available
- C) Too many valid outputs for open-ended tasks
- D) Models lack memory

✓ **Answer:** C

 **Explanation:** The slide emphasizes multiple valid answers problem (Page 39).

20. Which benchmark is specifically for competitive math?

- A) GLUE
- B) TruthfulQA
- C) MATH
- D) MedQA

✓ **Answer:** C

 **Explanation:** Mentioned multiple times on page 41 and 60.

21. What does HumanEval measure?

- A) Translation quality
- B) Coding execution correctness
- C) Summarization accuracy
- D) Model speed

✓ **Answer:** B

 **Explanation:** Used for evaluating code correctness (Page 45).

22. Which metric checks word insertions, deletions, and substitutions?

- A) BLEU
- B) METEOR
- C) WER
- D) TER

✓ **Answer:** C

 **Explanation:** Word Error Rate definition shown on Page 35.

23. Which one uses vector embeddings for semantic similarity?

- A) ROUGE
- B) BLEU
- C) METEOR
- D) BERTScore

✓ **Answer:** D

 **Explanation:** Mentioned under semantic similarity metrics (Page 46).

24. What does LLM-as-a-Judge risk introducing?

- A) Faster evaluation
- B) Self-preference bias
- C) Lower entropy
- D) Higher BLEU score

✓ **Answer:** B

 **Explanation:** The slide warns of self-preference bias (Page 52).

25. Why is comparative evaluation not suitable for yes/no questions?

- A) Because it wastes memory
- B) Because users don't know the correct answer
- C) Because ranking makes no sense when answer is binary
- D) Because yes/no answers are ambiguous

✓ **Answer:** C

 **Explanation:** Slide clearly mentions this (Page 50).

26. What new problem does Goodhart's Law warn about in LLM benchmarks?

- A) GPU failures
- B) Data contamination and overfitting
- C) Prompt injection
- D) Slower responses

✓ **Answer:** B

✎ **Explanation:** The slide mentions benchmarks may have been seen during training (Page 59).

27. What dataset is used to check if LLMs avoid false beliefs?

- A) MATH
- B) MedQA
- C) TruthfulQA
- D) OpenBookQA

✓ **Answer:** C

✎ **Explanation:** Mentioned explicitly on page 58.

28. How does SelfCheckGPT detect hallucinations?

- A) Using internet search
- B) Asking different users
- C) Checking answer consistency over multiple runs
- D) Using ref datasets

✓ **Answer:** C

✎ **Explanation:** Consistency implies factuality (Page 61).

29. Why is LLM-as-a-judge dangerous?

- A) Increases model size
- B) Slows down training
- C) Creates a filter bubble
- D) Requires extra RAM

✓ **Answer:** C

✎ **Explanation:** Models get trained to sound like the judge model (Page 52).

30. Which statement best summarizes the paper's evaluation philosophy?

- A) Only accuracy matters
- B) All models should be compared using BLEU

- C) Evaluation must be multi-dimensional and task-specific
- D) Perplexity is sufficient for all tasks

✓ **Answer:** C

 **Explanation:** Lecture repeatedly stresses domain-based and task-based evaluation (Pages 53–56).

31. In the RLHF pipeline diagram, what kind of data is used to fine-tune for “dialogue”?

- A) Internet crawl data
- B) Demonstration data
- C) Synthetic data
- D) Knowledge graph data

✓ **Answer:** B

 **Explanation:** The diagram explicitly labels “Demonstration data” as the source for supervised fine-tuning for dialogue (Page 13).

LLMS_ Alignment, Evaluation, Sy...

32. Which component in the RLHF flowchart outputs a scalar score for responses?

- A) SFT model
- B) Classification block
- C) Reward model
- D) Reinforcement learning block

✓ **Answer:** C

 **Explanation:** The Reward Model assigns a scalar score to each response based on human comparisons (Page 13).

LLMS_ Alignment, Evaluation, Sy...

33. What is the main risk of excessive censorship mentioned in preference fine-tuning?

- A) Model becomes biased
- B) Model becomes boring and less engaging
- C) Model requires more GPUs
- D) Model trains slower

✓ **Answer:** B

 **Explanation:** “Excessive censorship can make the model boring and drive users away” is stated directly (Page 10).

LLMS_ Alignment, Evaluation, Sy...

34. Which task category takes the **LARGEST** percentage in the task-distribution donut chart?

- A) Open QA
- B) Brainstorming
- C) Closed QA
- D) Generation

✓ **Answer:** D

 **Explanation:** In the donut chart (Page 9), “Generation” occupies **~45.6%**, the largest segment.

LLMS_ Alignment, Evaluation, Sy...

35. What is BLEU’s method of penalizing rare incorrect n-grams vs common ones?

- A) Same penalty for all
- B) Lower penalty for rare n-grams
- C) Higher penalty for rare n-grams
- D) Random penalty assignment

✓ **Answer:** C

 **Explanation:** Slide says rare n-grams like “buffalo buffalo” incur a higher penalty (Page 32).

LLMS_ Alignment, Evaluation, Sy...

36. Why does BLEU not “map well to human judgement”?

- A) It ignores sentence order
- B) It ignores meaning and structure
- C) It is too computationally expensive
- D) It is dataset dependent

✓ **Answer:** B

📝 **Explanation:** This is explicitly stated in the BLEU cons (Page 29).

LLMS_ Alignment, Evaluation, Sy...

37. Which metric is designed specifically for comparing sentences instead of corpora?

- A) BLEU
- B) TER
- C) METEOR
- D) ROUGE

✓ **Answer:** C

📝 **Explanation:** METEOR is explicitly said to compare sentences, unlike BLEU (Page 35).

LLMS_ Alignment, Evaluation, Sy...

38. Which metric penalizes outputs using syntactic tree structure?

- A) BLEU
- B) TER
- C) STM
- D) ROUGE-L

✓ **Answer:** C

📝 **Explanation:** Subtree Metric (STM) penalizes syntactic structure differences (Page 35).

LLMS_ Alignment, Evaluation, Sy...

39. Which statement best describes Cross Entropy?

- A) How random the dataset is
- B) How certain the model feels

- C) How surprised the model is when true distribution occurs
- D) How diverse vocab is

✓ **Answer:** C

 **Explanation:** Cross entropy is defined as model's surprise when world behaves like true distribution P (Page 43).

LLMS_ Alignment, Evaluation, Sy...

40. According to the lecture, what happens if $Q = P$ perfectly?

- A) KL divergence becomes infinite
- B) KL divergence becomes zero
- C) Cross entropy becomes zero
- D) Entropy becomes negative

✓ **Answer:** B

 **Explanation:** If distributions match, divergence equals zero (definition of KL on Page 43).

LLMS_ Alignment, Evaluation, Sy...

41. What does “domain-wise testing” mean for LLM evaluation?

- A) Testing on only one dataset
- B) Testing models separately per subject area
- C) Testing models only on math tasks
- D) Testing models using confusion matrices

✓ **Answer:** B

 **Explanation:** Page 53 states models are evaluated separately by domains like coding, QA, etc.

LLMS_ Alignment, Evaluation, Sy...

42. What does “exact evaluation” most likely imply in the context of the slides?

- A) Comparing semantic meaning
- B) Checking strict correctness

- C) Running LLM as judge
- D) Human feedback

✓ **Answer:** B

 **Explanation:** Page 54 lists “exact evaluation” separately from semantic and approximate metrics.

LLMS_ Alignment, Evaluation, Sy...

43. In programming evaluation, which metric is highlighted?

- A) ROUGE
- B) BLEU
- C) Functional correctness
- D) Entropy

✓ **Answer:** C

 **Explanation:** Code evaluation uses functional correctness like execution accuracy (Page 55).

LLMS_ Alignment, Evaluation, Sy...

44. Efficiency evaluation in the lecture considers which two main factors?

- A) Speed and accuracy
- B) Memory and GPU usage
- C) Time taken and memory usage
- D) Temperature and tokens

✓ **Answer:** C

 **Explanation:** Slide explicitly says efficiency = time + memory for SQL queries (Page 56).

LLMS_ Alignment, Evaluation, Sy...

45. Topical content evaluation often uses what format?

- A) Long essays
- B) Code problems
- C) MCQs
- D) Spoken QA

✓ **Answer:** C

📝 **Explanation:** “Topical content → MCQs” appears on page 57.

LLMS_ Alignment, Evaluation, Sy...

46. What special input variations are used in Natural Questions benchmark?

- A) With and without images
- B) With and without audio
- C) With and without Wikipedia pages
- D) With and without annotations

✓ **Answer:** C

📝 **Explanation:** Natural Questions is listed under two settings based on Wikipedia inclusion (Page 60).

LLMS_ Alignment, Evaluation, Sy...

47. Which benchmark focuses on medical reasoning?

- A) OpenBookQA
- B) MedQA
- C) LegalBench
- D) Story Cloze

✓ **Answer:** B

📝 **Explanation:** Medical benchmark listed explicitly (Page 60).

LLMS_ Alignment, Evaluation, Sy...

48. What does divergence in SelfCheckGPT responses indicate?

- A) High intelligence
- B) Overconfidence
- C) Hallucination
- D) Perfect model

✓ **Answer:** C

📝 **Explanation:** Divergence implies hallucination (Page 61).

49. Why is LLM-only evaluation considered risky?

- A) It increases compute cost
- B) It hides model weaknesses due to lack of architecture transparency
- C) It gives inconsistent outputs
- D) It reduces training time

 **Answer:** B

 **Explanation:** Slide mentions hiding architecture/training hides weaknesses (Page 40).